1. Compute the CDFs of R and θ in the Box–Muller algorithm. Hence show that their joint density on $[0, \infty) \times [0, 2\pi]$ is

$$f(r, \theta) = \frac{1}{2\pi} \exp\left(\frac{-r^2}{2}\right) r.$$

2. Consider sampling from the $N(0, \sigma^2)$-distribution via rejection sampling using a $N(0, \tau^2)$-proposal. (We leave aside the question of why one would want to do this.) Treat σ^2 as a fixed constant, and identify the values of τ^2 for which rejection sampling is valid. Which choice of τ^2 is optimal, and why?

3. Consider estimating the Gaussian tail probability $\mathbb{P}(X > \tau)$ using importance sampling with proposal density $\phi(x)$ given by the shifted exponential distribution,

$$\phi(x) = e^{-(x-\tau)} \text{ for } x \geq \tau.$$

   Show that the variance of the IS estimator obtained from $X_1, \ldots, X_N \overset{\text{iid}}{\sim} g(x)$ is

$$\text{Var}_\phi\left(\frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\{X_i > \tau\} \frac{\pi(X_i)}{\phi(X_i)}\right) = \frac{1}{N}\left[\frac{e^{1/4-\tau}}{2\pi}\int_\tau^\infty e^{-(x-1/2)^2} \mathrm{d}x - \mathbb{P}_\pi(X > \tau)^2\right].$$

4. Consider the centred and self-normalised IS estimate

$$\sqrt{N}(\widehat{I}_N^{\phi,\text{SN}} - \mathbb{E}[f(X)]) = \frac{\sqrt{N}\sum_{i=1}^N \frac{w(X_i)}{NZ}(f(X_i) - \mathbb{E}_\pi[f(X)])}{\sum_{j=1}^N \frac{w(X_j)}{NZ}} =: \frac{A_N}{B_N}.$$

   Check that the convergences stated in the lecture notes take place, i.e. that

$$\lim_{N\to\infty} A_N = N\left(0, \mathbb{E}_\pi\left[\frac{\pi(X_i)}{\phi(X_i)} f(X_i)^2\right] - \mathbb{E}_\pi\left[\frac{\pi(X_i)}{\phi(X_i)} f(X_i)\right]^2\right) \text{ in distribution,}$$

$$\lim_{N\to\infty} B_N = 1 \text{ almost surely.}$$

   What can you conclude about

$$\lim_{N\to\infty} \frac{1}{N}\sum_{j=1}^N w(X_j)?$$

5. Provide examples of the following, and show that they satisfy the stated criteria.

   (a) An irreducible Markov chain with a continuous state space. Your answer cannot be the mean-reverting random walk from the lecture slides.

   (b) A reducible Markov chain with a continuous state space with at least two stationary distributions.

   (c) A Markov chain with a continuous state space which has period > 1.

6. Consider a Markov chain on a compact state space Ω, and suppose its transition density can be written as
$$p(x, y) = (1 - \varepsilon)q(x, y) + \varepsilon\phi(y),$$
for some $\varepsilon \in (0, 1)$, where $q(x, y)$ is another transition density and $\phi(y)$ is a probability density on Ω. Show that the whole state space Ω is a small set.

7. Let $\varepsilon \ll 1$, and consider a Markov chain X_n on three states, $S = \{1, 2, 3\}$, with transition probability matrix
$$P = \begin{pmatrix} 1-\varepsilon & \varepsilon & 0 \\ 1/2 & 0 & 1/2 \\ 0 & \varepsilon & 1-\varepsilon \end{pmatrix}.$$
   (a) Find its unique stationary distribution.
   (b) Suppose $X_0 = 1$, and let K_ε be the random number of steps needed to cross from state 1 to state 3. Show that
$$\mathbb{P}(K_\varepsilon \leq k) \leq \frac{k\varepsilon}{4}.$$
   What can you conclude about how the number of steps needed for the chain to reach stationarity grows as $\varepsilon \to 0$?
   Hint: Consider the probability of starting at state 1 and crossing in two consecutive steps, and use a union bound.

8. Verify that detailed balance with respect to a distribution π implies that π is stationary.

9. Given a density $\pi(x)$, suppose (X_n) is a ϕ-irreducible, Harris-recurrent MCMC algorithm with the following dynamics.
   (1) Given X_{x-1}, propose $Y \sim q(X_{n-1}, \cdot)$, where $q(x, y)$ is a suitable proposal density.
   (2) Set $X_n = Y$ with probability
$$\frac{\pi(Y)q(Y, X_{n-1})}{\pi(Y)q(Y, X_{n-1}) + \pi(X_{n-1})q(X_{n-1}, Y)}.$$
   (3) Else set $X_n = X_{n-1}$.

   This chain is known as Barker's algorithm.
   (a) Assuming that $q(x, \cdot)$ has the same support as $\pi(\cdot)$, show that Barker's algorithm has stationary density π.
   (b) Show that the acceptance probability of the Metropolis–Hastings algorithm with the same proposal distribution is at least as high as that of the Barker algorithm.